

Thus, another linearly independent solution is $u_2(x) = \cos kx/x$.

(ii) A straightforward calculation gives the Wronskian of u_1 and u_2 as

$$W(x) = -k/x^2.$$

Now, a particular integral of the given equation is [from Eq.(5.32)]

$$u_p(x) = g(x) \frac{\sin kx}{x} + h(x) \frac{\cos kx}{x}$$

where

$$g(x) = - \int_{x_0}^x \frac{u_2(x') f(x')}{W(x')} dx' \\ = \frac{A}{k} \int_{x_0}^x x' \cos kx' \sin \beta x' dx'$$

and

$$h(x) = - \int_{x_0}^x \frac{u_1(x') f(x')}{W(x')} dx' \\ = - \frac{A}{k} \int_{x_0}^x x' \sin kx' \sin \beta x' dx'.$$

5.3 Power series method

We write Eq.(5.17) as

$$u''(z) + p(z)u'(z) + q(z)u(z) = 0 \quad (5.35)$$

with the complex variable z as the independent variable. (This is convenient for theoretical discussion even though z is real for most physical applications.) A point z_0 in the complex z -plane is called an 'ordinary point' of the differential equation (5.35) if $p(z)$ and $q(z)$ are analytic at z_0 ; otherwise, it is a 'singular point'. A singular point z_0 is called a 'regular singular point' if, at z_0 , $p(z)$ has a pole of order not greater than one and $q(z)$ has a pole of order not greater than two; otherwise, it is called an 'irregular singular point'.

We now state and prove the two basic theorems relating to power series solutions of Eq.(5.35).

Theorem 5.2 If z_0 is an ordinary point of the differential equation (5.35), it admits, in a neighbourhood of z_0 , two linearly independent power series solutions of the form

$$u(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n. \quad (5.36)$$

Theorem 5.3 If z_0 is a regular singular point of Eq.(5.35), it admits, in a neighbourhood of z_0 , at least one solution of the form ✓

$$u_1(z) = (z - z_0)^{r_1} \sum_{n=0}^{\infty} c_n (z - z_0)^n \quad (5.37)$$

(where r_1 is, in general, a complex number). Another linearly independent solution is of the form

$$u_2(z) = (z - z_0)^{r_2} \sum_{n=0}^{\infty} d_n (z - z_0)^n + K_1 u_1(z) \ln(z - z_0) \quad \checkmark \quad (5.38)$$

where K_1 is a constant and $d_0 = 0$ for the special case $r_2 = r_1$.

In special cases, the constant K may be zero so that the equation admits two linearly independent solutions of the form (5.37). Note that when $K \neq 0$, $u_2(z)$ is singular at $z = z_0$.

Proof of Theorem 5.2 We take $z_0 = 0$ for simplicity. (This is equivalent to a change of variable $z' = z - z_0$ and replacing z' by z .) Since $p(z)$ and $q(z)$ are analytic at $z = 0$, we can write, for some R ,

$$p(z) = \sum_{n=0}^{\infty} a_n z^n, \quad q(z) = \sum_{n=0}^{\infty} b_n z^n, \quad |z| < R. \quad (5.39)$$

Let

$$u(z) = \sum_{n=0}^{\infty} c_n z^n. \quad (5.40)$$

We must show that the coefficients c_n can be found such that the series (5.40) converges in some domain about the origin, and the resulting function $u(z)$ satisfies Eq.(5.35).

Substituting Eqs.(5.39) and (5.40) in Eq.(5.35) and equating to zero the coefficient of z^{n-2} , we obtain

$$-n(n-1)c_n = \sum_{k=1}^{n-1} k a_{n-1-k} c_k + \sum_{m=0}^{n-2} b_{n-2-m} c_m. \quad (5.41)$$

By Cauchy's integral formula, we have

$$a_n = \frac{1}{2\pi i} \oint_C \frac{p(z)}{z^{n+1}} dz, \quad b_n = \frac{1}{2\pi i} \oint_C \frac{q(z)}{z^{n+1}} dz \quad (5.42)$$

where C is a circle of radius $r < R$ around the origin. Let

$$|p(z)| \leq L, \quad |q(z)| \leq M, \quad \text{for } z \in C.$$

Equation (5.42) then gives

$$|a_n| \leq \frac{1}{2\pi} \frac{L}{r^{n+1}} 2\pi r = \frac{L}{r^n}, \quad |b_n| \leq \frac{M}{r^n}.$$

Let $K = \max(L, M)$. Then

$$|a_n| \leq Kr^{-n}, \quad |b_n| \leq Kr^{-n}. \quad (5.43)$$

We shall now prove that

$$|c_n| \leq Bs^{-n}, \quad \text{for some positive constants } B \text{ and } s. \quad (5.44)$$

The proof proceeds by induction. For $n = 0, 1$, Eq.(5.44) gives

$$|c_0| \leq B, \quad |c_1| \leq Bs^{-1}. \quad (5.45)$$

The coefficients c_0 and c_1 will ultimately be left to be chosen freely; we choose them so as to satisfy Eq.(5.45) for some B and s . Now assume that Eq.(5.44) holds for $c_0, c_1, \dots, c_{n-1}$. Equation (5.41) then gives

$$\begin{aligned} n(n-1)|c_n| &\leq \left| \sum_{k=1}^{n-1} ka_{n-1-k}c_k \right| + \left| \sum_{m=0}^{n-2} b_{n-2-m}c_m \right| \leq \sum_{k=1}^{n-1} k|a_{n-1-k}||c_k| + \sum_{m=0}^{n-2} |b_{n-2-m}||c_m| \\ &\leq BK \left[\sum_{k=1}^{n-1} kr^{-n+1+k}s^{-k} + \sum_{m=0}^{n-2} r^{-n+2+m}s^{-m} \right]. \end{aligned} \quad (5.46)$$

Take $s < r$. Then Eq.(5.46) gives

$$\begin{aligned} n(n-1)|c_n| &\leq BK \left[\sum_{k=1}^{n-1} ks^{-n+1} + \sum_{m=0}^{n-2} s^{-n+2} \right] \\ &= BK \left[\frac{n(n-1)}{2}s^{-n+1} + (n-1)s^{-n+2} \right]. \end{aligned}$$

It follows that

$$|c_n| \leq BKs^{-n} \left[\frac{s}{2} + \frac{s^2}{n} \right] \leq BKs^{-n} \left(\frac{s}{2} + s^2 \right). \quad (5.47)$$

Choose s sufficiently small so that

$$K \left(\frac{s}{2} + s^2 \right) \leq 1. \quad (5.48)$$

Equations (5.47) and (5.48) then give Eq.(5.44).

Now the series $\sum |c_n||z|^n$ is term by term less than the series

$$B \sum_n (|z|/s)^n$$

which is a geometric series converging for $|z| < s$. It follows that the series (5.40) is absolutely convergent for $|z| < s$.

The recurrence relation (5.41) determines all coefficients c_n in terms of c_0 and c_1 . We can, therefore, write

$$u(z) = c_0 u_1(z) + c_1 u_2(z) \quad (5.49)$$

where u_1 and u_2 are infinite series (each converging for $|z| < s$; this is because u_1 and u_2 are special cases of u corresponding to $c_1 = 0$ and $c_0 = 0$ respectively) solutions of Eq.(5.35). Moreover,

$$\begin{aligned} u_1(0) &= 1, & u_1'(0) &= 0 \\ u_2(0) &= 0, & u_2'(0) &= 1. \end{aligned} \quad (5.50)$$

To see that the solutions u_1 and u_2 are linearly independent, let

$$au_1(z) + bu_2(z) = 0.$$

Using Eq.(5.50), it is easily seen that we must have $a = b = 0$. (QED)

Remarks

1. From Eq.(5.50) we can easily deduce that the initial value problem consisting of Eq.(5.35) with the initial conditions

$$u(0) = A, \quad u'(0) = B \quad (5.51)$$

has the unique solution

$$u(z) = Au_1(z) + Bu_2(z). \quad (5.52)$$

2. A general solution of the inhomogeneous equation (5.16) around an ordinary point can be directly obtained by power series method if the inhomogeneous term admits a series expansion around the point z_0 (see Example 5.2 below).
3. When $z(=x)$ is real and Eq.(5.35) has two or more singular points on the real axis, the range of values of x for which the solutions obtained above are valid, is generally the domain between two consecutive singular points of the equation.

Example 5.2 Obtain a solution of the differential equation

$$u'' + (x+2)u' + u = 1 + 4x + x^2$$

as a power series in $(x+2)$.

Solution We change the independent variable from x to $y = x+2$. The equation then becomes

$$u'' + yu' + u = y^2 - 3.$$

As $y=0$ is an ordinary point of the differential equation, we can obtain a solution of the form

$$u(y) = \sum_{n=0}^{\infty} b_n y^n.$$

Substituting in the differential equation, we obtain

$$\sum_{n=2}^{\infty} n(n-1)b_n y^{n-2} + y \sum_{n=1}^{\infty} n b_n y^{n-1} + \sum_{n=0}^{\infty} b_n y^n = y^2 - 3$$

or,

$$\left[2b_2 + 6b_3y + 12b_4y^2 + \sum_{n=5}^{\infty} n(n-1)b_n y^{n-2} \right] + \left[b_1y + 2b_2y^2 + \sum_{n=3}^{\infty} nb_n y^n \right] + \left[b_0 + b_1y + b_2y^2 + \sum_{n=3}^{\infty} b_n y^n \right] = y^2 - 3.$$

Equating coefficients of various powers of y gives

$$y^0 : 2b_2 + b_0 = -3$$

$$y^1 : 6b_3 + 2b_1 = 0$$

$$y^2 : 12b_4 + 3b_2 = 1$$

$$y^n (n > 2) : (n+2)b_{n+2} + b_n = 0$$

The required series is thus given by

$$u(x) = \sum_{n=0}^{\infty} b_n (x+2)^n$$

where $b_2 = -(b_0 + 3)/2$, $b_3 = -b_1/3$, $b_4 = -(11 + 3b_0)/24$ and $b_{n+2} = -b_n/(n+2)$ ($n > 2$), etc. Here b_0, b_1 are arbitrary.

Since the given equation has no singular point in the real line, we expect, from Remark 3 above, the solution to be valid for all real values of x . This is, indeed, so as can be seen from the relation

$$\lim_{n \rightarrow \infty} \frac{b_{n+2}}{b_n} = 0$$

implying convergence of the series for all real values of x .

Proof of Theorem 5.3 Again we take $z_0 = 0$. Instead of Eqs.(5.39), we are now given $P(z) = zp(z)$ and $Q(z) = z^2q(z)$ to be analytic at the origin; we can, therefore write

$$P(z) = \sum_{n=0}^{\infty} p_n z^n; \quad Q(z) = \sum_{n=0}^{\infty} q_n z^n, \quad |z| < R. \quad (5.53a,b)$$

Note that since the functions $p(z)$ and $q(z)$ are real for real values of z (we started with the differential equation (5.17) in which everything was real), the coefficients p_n and q_n in Eqs.(5.53) must be real. We now try a solution of the form

$$u(z) = \sum_{n=0}^{\infty} c_n z^{n+r}, \quad c_0 \neq 0. \quad (5.54)$$

Substituting the expansions (5.53) and (5.54) in Eq.(5.35), we have

$$\sum_{n=0}^{\infty} (n+r)(n+r-1)c_n z^{n+r-2} + \left(\sum_{k=0}^{\infty} p_k z^{k-1} \right) \left(\sum_{m=0}^{\infty} (m+r)c_m z^{m+r-1} \right) + \left(\sum_{k=0}^{\infty} q_k z^{k-2} \right) \left(\sum_{m=0}^{\infty} c_m z^{m+r} \right) = 0. \quad (5.55)$$

Equating the coefficient of the lowest power of z (i.e. z^{r-2}) to zero, we get

$$[r(r-1) + p_0 r + q_0] c_0 = 0.$$

Since $c_0 \neq 0$, this gives the so-called 'indicial equation'

$$E(r) \equiv r(r-1) + p_0 r + q_0 = 0 \quad \checkmark \quad (5.56)$$

which has the solutions

$$r = \frac{1}{2} \left[(1 - p_0) \pm \sqrt{(p_0 - 1)^2 - 4q_0} \right]. \quad (5.57)$$

Calling these solutions r_1, r_2 , we note that

$$\operatorname{Re}(r_1 - r_2) \geq 0, \quad \operatorname{Re} r_1 \geq \frac{1}{2} \operatorname{Re}(1 - p_0). \quad (5.58)$$

We shall now show that, (i) when $r_1 - r_2$ is not an integer, convergent series solutions with both values of r exist and are linearly independent, and (ii) if $r_1 - r_2$ is an integer, a convergent solution with $r = r_1$ exists; a second linearly independent solution can then be found by variation of parameters.
 OK! Fuchs's Theorem

Case (i): $r_1 - r_2$ not an integer

In the following, we have $r = r_1$ or r_2 so that Eq.(5.56) holds. Equating the coefficient of z^{n+r-2} in Eq.(5.55) to zero, we have

$$(n+r)(n+r-1)c_n + \sum_{m=0}^n [(m+r)p_{n-m} + q_{n-m}] c_m = 0$$

or

$$E(n+r)c_n + \sum_{m=0}^{n-1} [(m+r)p_{n-m} + q_{n-m}] c_m = 0, \quad n = 1, 2, \dots \quad (5.59)$$

This recursion relation determines all the coefficients $c_1, c_2, \dots$ in terms of c_0 . The convergence proof proceeds along the same lines as for Theorem 5.2. Let

$$|P(z)| < K_1, \quad |Q(z)| < K_2 \quad \text{for} \quad |z| < R. \quad (5.60)$$

We have

$$p_n = \frac{1}{2\pi i} \oint_C \frac{P(z) dz}{z^{n+1}}, \quad q_n = \frac{1}{2\pi i} \oint_C \frac{Q(z) dz}{z^{n+1}} \quad (5.61)$$

where C is a circle of radius $a < R$. This gives [see Eq.(5.43)]

$$|p_n| \leq Ka^{-n}, \quad |q_n| \leq Ka^{-n} \quad (5.62)$$

where $K = \max(K_1, K_2)$. Equation (5.59) then gives

$$|E(n+r)| |c_n| \leq \sum_{m=0}^{n-1} (m+|r|+1) Ka^{-n+m} |c_m|. \quad (5.63)$$

Now, using Eq.(5.56), we have

$$\begin{aligned} E(n+r) &= (n+r)(n+r-1) + p_0(n+r) + q_0 \\ &= n^2 + n(2r-1+p_0). \end{aligned} \quad (5.64)$$

For $r = r_1$, we have, by Eq.(5.58)

$$\operatorname{Re} [(2r-1) + p_0] \geq 0.$$

Equation (5.64) then gives

$$|E(n+r_1)| \geq n^2.$$

Equation (5.63) gives

$$|c_n| \leq \sum_{m=0}^{n-1} (m+|r|+1) Kn^{-2} a^{-n+m} |c_m|.$$

Now,

$$K \left[\frac{m}{n} + \frac{|r|+1}{n} \right] \leq K(|r|+2) = L, \text{ say.}$$

We have, therefore,

$$|c_n| \leq Ln^{-1} \sum_{m=0}^{n-1} a^{-n+m} |c_m|. \quad (5.66)$$

We shall now prove that

$$|c_n| \leq \left(\frac{L}{a} \right)^n |c_0|. \quad (5.67)$$

The proof proceeds by induction. Equation (5.67) clearly holds for $n=1$. Now assume that it holds for $n=1, 2, \dots, (N-1)$. Then Eq.(5.66) gives

$$\begin{aligned} |c_N| &\leq LN^{-1} \sum_{m=0}^{N-1} a^{-N+m} \left(\frac{L}{a} \right)^m |c_0| \\ &= \frac{L|c_0|}{Na^N} \sum_{m=0}^{N-1} L^m. \end{aligned} \quad (5.68)$$

We can assume, without loss of generality, that $L > 1$. Then

$$\sum_{m=0}^{N-1} L^m < NL^{N-1}.$$

Equation (5.68) now gives

$$|c_N| \leq \left(\frac{L}{a}\right)^N |c_0|$$

which proves Eq.(5.67). The convergence of the series (5.54) (with $r = r_1$) for $|z| < aL^{-1}$ now follows as before.

For $r = r_2$, the treatment after Eq.(5.64) needs to be modified. Note that $E(n + r_2)$ cannot vanish. Now

$$2r_2 - 1 + p_0 = -\sqrt{(p_0 - 1)^2 - 4q_0} = A + iB, \text{ say.}$$

Instead of Eq.(5.65), we now have

$$|E(n + r_2)| \geq |(n^2 - n|A|)|.$$

Proceeding as before, we can now prove that [see Eq.(5.67)]

$$|c_n| \leq \left(\frac{L}{a}\right)^n \frac{n^2}{|(n^2 - n|A|)|} |c_0|.$$

Clearly, for large n , the behaviour of the coefficients c_n is the same as for $r = r_1$ and once again, we have a convergent series solution.

To see that the solutions with $r = r_1$ and $r = r_2$ are linearly independent, let the two solutions be

$$u_1(z) = \sum_{n=0}^{\infty} c_n z^{n+r_1}, \quad u_2(z) = \sum_{n=0}^{\infty} d_n z^{n+r_2}.$$

$r_1 - r_2$ not an integer!

The Wronskian of the two solutions is

$$\begin{aligned} W(u_1, u_2)(z) &= (u_1 u_2' - u_1' u_2)(z) \\ &= \sum_{m,n} c_m d_n (n + r_2 - m - r_1) z^{n+m+r_1+r_2-1}. \end{aligned}$$

Since $r_1 - r_2$ is not an integer, the Wronskian is clearly nonzero and the two solutions are linearly independent.

Case (ii): $r_1 - r_2 = \text{integer} = N$, say.

Note, from Eq.(5.58), that $N \geq 0$. In this case, $E(N + r_2) = 0$ and the construction of c_n 's from c_0 by the recursion formula (5.59) for the solution with $r = r_2$ cannot proceed.

The method described above, therefore, provides only one solution $u_1(z)$ corresponding $r = r_1$. To construct another solution, we follow the method of variation of parameter. Writing $u_2(z) = K(z)u_1(z)$, Eq.(5.21) gives

$$\frac{dK(z)}{dz} = \frac{\text{constant}}{[u_1(z)]^2} \exp \left[- \int_0^z p(z') dz' \right] \quad (5.6)$$

where the integral is along any path joining 0 and z , and lying in the domain of analyticity of the function $p(z)$. Now Eq.(5.53a) gives

$$\exp \left[- \int_0^z p(z') dz' \right] = \exp \left[-p_0 \ln z - \sum_{n=1}^{\infty} p_n n^{-1} z^n \right]$$

Equation (5.69) now gives

$$\frac{dK}{dz} = z^{-p_0-2r_1} G(z)$$

where, the function

$$G(z) = (\text{constant}) \exp \left[- \sum_{n=1}^{\infty} n^{-1} p_n z^n \right] \left[\sum_{n=0}^{\infty} c_n z^n \right]^{-2} \quad (5.70)$$

being analytic at $z = 0$, can be expanded as

$$G(z) = \sum_{n=0}^{\infty} g_n z^n. \quad (5.71)$$

Now from Eq.(5.57), we have

$$p_0 + 2r_1 = p_0 + (r_1 + r_2) + (r_1 - r_2) = p_0 + (1 - p_0) + N = N + 1.$$

We have, therefore,

$$\frac{dK}{dz} = \sum_{n=0}^{\infty} g_n z^{n-N-1}$$

which gives, ignoring an additive constant in K ,

$$K(z) = g_N \ln z + \sum_{n \neq N} g_n (n - N)^{-1} z^{n-N} \quad (5.72)$$

which, in turn, gives $u_2(z)$ in the form of Eq.(5.38). ✓

(QED)

Example 5.3 Obtain two linearly independent solutions of the equation

$$x^2 u'' + 4xu' + (x^2 + 2)u = 0.$$